Negative Imaginary Stability Result Allowing Purely Imaginary Poles in Both the Interconnected Systems

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***Abstract*—This letter extends the class of systems to which negative imaginary (NI) stability result can be (directly) applied. The existing closed-loop stability results for NI interconnections require at least one of the systems to satisfy *strictly* negative imaginary (SNI) properties and hence do not allow purely imaginary poles in both the systems. In contrast to this, a set of necessary and sufficient conditions is derived in this brief for the internal stability of two NI systems in a positive feedback loop with possible poles on the imaginary axis excluding the origin. This proposed result also captures all the existing closed-loop stability results of NI systems, without poles at the origin, as special cases. Illustrative numerical examples are studied to show the usefulness of this development.**

***Index Terms*—Negative imaginary system, positive feedback, DC loop gain, internal stability.**

I. Introduction

NEGATIVE imaginary (NI) systems property and the associated robust stability result were introduced in [1] motivated by the 'positive position feedback control' of highly resonant flexible structures having collocated force actuators and position sensors. In SISO setting, the term NI signifies a class of systems having negative imaginary frequency response, that is, their Nyquist plots reside in the union of third and fourth quadrants of the complex plane for all $\omega \in [0, \infty)$ [1], [2]. NI properties are quite common in mechanical [1]–[4], electrical [5], [6], electromechanical [7] and mechatronic [8] systems. NI theory finds potential applications in vibration control of lightly-damped flexible structures [1], [3], cantilever beams [4], large space structures with free-body motion and robotic manipulators [9], in nano-positioning applications [8], in control of large vehicle platoons [10], in designing stabilizing and tracking controllers for complex nonlinear systems [11]–[13], etc. The NI theory has become appealing due to its simple internal stability condition that depends on the loop gain only at zero frequency, and since it offers a stand-alone robust control analysis and synthesis framework [1], [14]–[18]. NI theory has recently been extended to capture improper and non-rational systems [19]–[21], and also to discrete-time LTI systems [22]–[24].

Initially, in [1], internal stability of a positive feedback interconnection containing a stable NI (say, $M(s)$) and an SNI (say, $N(s)$) system (see Fig. 1) was proved. A necessary and sufficient condition $\lambda_{\max}[N(0)M(0)] < 1$ (well-known as the *DC loop gain condition* in the NI literature) was derived to guarantee internal stability of a NI–SNI interconnection under the technical assumptions $M(\infty)N(\infty) = 0$ and $N(\infty) \geq 0$. Subsequently, this result was extended in [3] where the (non-strict) NI system was allowed to accept non-repeated poles on the $j\omega$-axis excluding the origin. This stability result was subsequently specialized in [25] when the NI system was constrained to contain only $j\omega$-axis poles (referred to as the 'lossless NI' systems). Later, in [26], the feedback stability results of a NI–SNI interconnection were generalized by removing the earlier restrictions: $M(\infty)N(\infty) = 0$ and $N(\infty) \geq 0$. Meanwhile, NI definition was also extended to systems having up to two poles at the origin [9], [27]. The associated stability results are also available in the literature. Note that, for internal stability, all the current results in the NI literature must require one of the systems in the interconnection to be SNI. However, in some practical applications (e.g., large space structures, active suspension systems, shock absorbers [2]), we encounter the stability problems of interconnected NI systems, where both the systems may contain poles on the $j\omega$-axis. This letter aims to develop a necessary and sufficient internal stability condition for two NI systems connected in a positive feedback loop that would allow both the systems to have simple poles on the $j\omega$-axis (referred to as *marginally stable* NI systems in this letter) excluding the origin. This result would, therefore, require none of the systems to be SNI or even stable NI. The stability theorem proposed in this letter eventually encompasses all the existing internal stability results of an NI–SNI interconnection, except when the NI system has poles at the origin. It may also be noted that the internal stability of two marginally stable NI

Manuscript received January 22, 2021; revised April 10, 2021; accepted April 20, 2021. Date of publication May 5, 2021; date of current version June 25, 2021. Recommended by Senior Editor L. Menini. *(Corresponding author: Arnab Dey.)*

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Digital Object Identifier 10.1109/LCSYS.2021.3077862

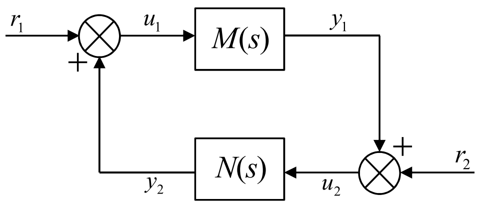

Fig. 1. Positive feedback interconnection of two NI systems.

systems cannot be captured by the Small-gain theorem or the Passivity theorem [28]. Therefore, we believe that the proposed result will also contribute to the robust control literature.

Notation: The notation is standard throughout. The set of all natural numbers (excluding 0) is denoted as $\mathbb{N} = \{1, 2, 3, \dots, \}$. $A^\top$, A^* and $\bar{A}$ denote the transpose, the complex conjugate transpose and the complex conjugate of a matrix A. A^{-*} and $A^{-\top}$ represent shorthand for $(A^{-1})^*$ and $(A^{-1})^\top$ respectively. $\mathscr{R}^{m\times n}$ and $\mathscr{RH}_\infty^{m\times n}$ denote respectively the sets of all proper, real, rational, and all proper, real, rational and asymptotically stable transfer function matrices, both of dimensions $(m \times n)$. For a transfer function matrix $M(s)$, $M(j\omega)^* = M(-j\omega)^\top$. The real-Hermitian and imaginary-Hermitian frequency response parts are given by $\frac{1}{2}[M(j\omega) + M(j\omega)^*]$ and $\frac{1}{2j}[M(j\omega) - M(j\omega)^*]$ respectively. The $\mathscr{L}_2$-adjoint of a transfer function matrix $M(s)$ is expressed as $M^\sim(s) = M(-s)^\top$. $\left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right]$ denotes a state-space realization of a real, rational, proper transfer function matrix $M(s) = D + C(sI - A)^{-1}B$. The symbol $\emptyset$ denotes an empty set. The relational operator $\not\equiv$ is used to mean 'not identically equal to'.

II. Technical Preliminaries

Here, we provide precise background information on NI and SNI systems and their internal stability results that exist in the literature. This is followed by some new results on the algebraic connections between the rank deficiency of a system transfer function matrix (due to the presence of transmission zeros on the $j\omega$-axis) and that of its imaginary-Hermitian part.

Definition 1 (NI System[1] [3], [26]): Let $M(s)$ be the real, rational and proper transfer function matrix of a finite-dimensional, square and causal system without any poles in the open right-half plane. Then, $M(s)$ is said to be NI if

- $j[M(j\omega) - M(j\omega)^*] \geq 0$ for all $\omega \in (0, \infty)$ except the values of ω where $s = j\omega$ is a pole of $M(s)$;
- If $s = j\omega_0$ with $\omega_0 \in (0, \infty)$ is a pole of $M(s)$, then it is at most a simple pole and the residue matrix $\lim_{s\to j\omega_0} (s - j\omega_0)jM(s)$ is Hermitian and positive semidefinite.

Definition 2 (SNI System [1], [26]): Let $M(s)$ be the real, rational and proper transfer function matrix of a finite-dimensional, square and causal system. Then, $M(s)$ is said to be strictly negative imaginary if $M(s)$ has no poles in $\{s \in \mathbb{C} : \Re[s] \geq 0\}$ and $j[M(j\omega) - M(j\omega)^*] > 0$ for all $\omega \in (0, \infty)$.

[1]This definition can be extended to account for NI systems containing up to two poles at the origin and also at infinity (refer to [9], [26]).

The following lemma, referred to as the NI Lemma, provides a state-space characterization for NI systems.

Lemma 1 (NI Lemma [3], [26]): Let $M(s)$ be the real, rational, proper transfer function matrix of a square system having a minimal state-space realization $\left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right]$. Then, $M(s)$ is NI without poles at the origin if and only if $\det[A] \neq 0$, $D = D^\top$ and there exists a real matrix $Y = Y^\top > 0$ such that

$$AY + YA^\top \leq 0 \qquad \text{and} \qquad B + AYC^\top = 0. \tag{1}$$

Let us now recall the feedback stability conditions of a NI–SNI interconnection given in [26, Th. 9] which removes all the previous restrictions on the gains of the systems at infinite frequency.

Theorem 1 (NI–SNI stability theorem [26]): Let $M(s)$ be an NI system without poles at the origin and $N(s)$ be an SNI system. Then, the positive feedback interconnection of $M(s)$ and $N(s)$, shown in Fig. 1, is internally stable if and only if

$$\det[I - M(\infty)N(\infty)] \neq 0, \tag{2a}$$

$$\lambda_{\max}\left[(I - M(\infty)N(\infty))^{-1}(M(\infty)N(0) - I)\right] < 0, \tag{2b}$$

$$\lambda_{\max}\left[(I - N(0)M(\infty))^{-1}(N(0)M(0) - I)\right] < 0. \tag{2c}$$

If the previous assumptions $M(\infty)N(\infty) = 0$ and $N(\infty) \geq 0$ are imposed, then the stability criteria (2a)–(2c) reduces to the well-known DC gain condition $\lambda_{\max}[N(0)M(0)] < 1$ [26, Corollary 12], [3].

Below, we establish a relationship between the transmission zeros of an NI transfer function matrix on the imaginary axis and the rank deficiency of its imaginary-Hermitian part at that frequency. This result will be used later in Section III to establish the main result of this letter.

Lemma 2: Let $G(s) \in \mathscr{R}^{m\times m}$ be an NI system with full normal rank. Suppose $s = j\omega_z$ with $\omega_z \in (0, \infty)$ being a transmission zero of $G(s)$ but not a pole. Then, $\det[G(j\omega_z) - G(j\omega_z)^*] = 0$.

Proof: As $s = j\omega_z$, $\omega_z \in (0, \infty)$, is a transmission zero of $G(s)$, there exists a non-zero vector $x \in \mathbb{C}^m$ such that $G(j\omega_z)x = 0$. This then implies $x^*G(j\omega_z)x = 0 \Rightarrow x^*\frac{1}{2j}[G(j\omega_z) - G(j\omega_z)^*]x = 0 \Leftrightarrow x^*j[G(j\omega_z) - G(j\omega_z)^*]x = 0$. For convenience, let $Z = j[G(j\omega_z) - G(j\omega_z)^*]$. Now, $Z = Z^* \geq 0$ as $G(s)$ is NI and $s = j\omega_z$ is not a pole of $G(s)$. Since $Z \geq 0$, there exists a unique matrix square root $Z^{\frac{1}{2}} \geq 0$ (see [29, p. 406]). Therefore, we have $x^*Z^{\frac{1}{2}}Z^{\frac{1}{2}}x = 0 \Leftrightarrow y^*y = 0$ by denoting $y = Z^{\frac{1}{2}}x$, which in turn implies $y = 0$. Hence, $\det[Z] = 0$ as $x \neq 0$. This completes the proof. ■

Lemma 3: Let $G(s) \in \mathscr{R}^{m\times m}$ be an NI system. Then, $j[G(j\omega_z) - G(j\omega_z)^*] > 0$ with $\omega_z \in (0, \infty)$ implies $\det[G(j\omega_z)] \neq 0$.

Proof: Suppose via contradiction that $\det[G(j\omega_z)] = 0$. Then, there exists a non-zero vector $x \in \mathbb{C}^m$ such that $G(j\omega_z)x = 0$ which ultimately implies $x^*[j\{G(j\omega_z) - G(j\omega_z)^*\}]x = 0$, as shown in the proof of Lemma 2. But, the result violates [29, Chapter 7] the supposition that $j[G(j\omega_z) - G(j\omega_z)^*] > 0$. Hence, there does not exist any non-zero $x \in \mathbb{C}^m$ such that $G(j\omega_z)x = 0$, that is, $\det[G(j\omega_z)] \neq 0$. This completes the proof. ■

III. Internal Stability of NI Systems Having $j\omega$-Axis Poles

This section contains the main contribution of this letter that offers an extended internal stability result for interconnected NI systems (Fig. 1) with possible poles on the $j\omega$-axis excluding the origin. This eliminates the restriction that to ensure internal stability of an NI interconnection, one of the systems must be SNI. To prove the main stability theorem, we need to establish first an essential prerequisite technical result presented in the following Lemma.

Lemma 4: Let $M(s)$ and $N(s)$ be two NI systems with possible poles on the $j\omega$-axis excluding the origin. Let $\Omega_m = \{\omega : \omega \in (0,\infty),\ s = j\omega$ is a pole of $M(s)\}$ and $\Omega_n = \{\omega : \omega \in (0,\infty),\ s = j\omega$ is a pole of $N(s)\}$ such that $\Omega_m \cap \Omega_n = \emptyset$. Let $M(s)N(s)$ have no pole-zero cancellation on the $j\omega$-axis for any $\omega \in (0,\infty)$ and

$$j\big[N(j\omega_k) - N(j\omega_k)^*\big] > 0 \quad \forall \omega_k \in \Omega_m, \tag{3}$$
$$j\big[M(j\omega_l) - M(j\omega_l)^*\big] > 0 \quad \forall \omega_l \in \Omega_n. \tag{4}$$

Suppose finally that *either* $\nexists$ any $\omega \in (0,\infty)\backslash(\Omega_m \cup \Omega_n)$ such that $\det[M(j\omega) - M(j\omega)^*] = 0$ and $\det[N(j\omega) - N(j\omega)^*] = 0$ when both $[M(s) - M^{\sim}(s)]$ and $[N(s) - N^{\sim}(s)]$ have full normal rank; *or* $\det[I - M(j\omega_0)N(j\omega_0)] \neq 0$ where $\omega_0 \in \{\omega \in (0,\infty)\backslash(\Omega_m \cup \Omega_n) : \det[N(j\omega) - N(j\omega)^*] = 0$ (respectively $\det[M(j\omega) - M(j\omega)^*] = 0)\}$ when $[M(s) - M^{\sim}(s)]$ (respectively $[N(s) - N^{\sim}(s)]$) loses its full normal rank. Then, $[I - M(s)N(s)]$ does not have any transmission zero at $s = j\omega$ for any $\omega \in (0,\infty)$.

Proof: The proof has been divided into four cases. Case I proves the lemma when both the systems are marginally stable NI without poles at the origin. Case II and Case III apply to the scenarios where one of the systems in the interconnection is stable NI. Case IV deals with an NI interconnection where both the systems are stable.

Case I: Suppose $M(s)$ and $N(s)$ have respectively $K \in \mathbb{N}$ and $L \in \mathbb{N}$ non-repeated poles on $j\omega$-axis for $\omega \in (0,\infty)$. Define the sets $\underline{K} = \{1, 2, \dots, K\}$ and $\underline{L} = \{1, 2, \dots, L\}$. Since both $M(s)$ and $N(s)$ do not have any poles in the open right-half plane or at the origin and via supposition, $M(s)N(s)$ does not have any pole-zero cancellation on the $j\omega$-axis for $\omega \in (0,\infty)$, it can be easily seen that $M(s)N(s)$ does not cause any pole-zero cancellation in the entire closed right-half plane. Case I is proved in two parts: Part 1 proves the result for the set of frequencies $\omega \in (\Omega_m \cup \Omega_n)$, while Part 2 establishes the desired result for all $\omega \in (0,\infty)\backslash(\Omega_m \cup \Omega_n)$. Note further that the assumptions of this lemma ensure that $\det[M(j\omega) - M(j\omega)^*] \not\equiv 0$ and $\det[N(j\omega) - N(j\omega)^*] \not\equiv 0\ \forall \omega \in (0,\infty)$ which imply that $[M(s) - M^{\sim}(s)]$ and $[N(s) - N^{\sim}(s)]$ have full normal rank.

Part 1 ($\omega \in (\Omega_m \cup \Omega_n)$): Since $s = j\omega_k$ for each $k \in \underline{K}$ is a simple pole of $M(s)$, $M(s)$ can be factorized as $M_k(s) + \dfrac{A_k}{s - j\omega_k}$ with $M_k(s)$ being analytic in the neighborhood of $s = j\omega_k$ and $A_k = \lim_{s \to j\omega_k} (s - j\omega_k)M(s)$. Now the residue property of NI systems (condition 3 of Definition 1) gives $K_k = \lim_{s \to j\omega_k} j(s - j\omega_k)M(s) = K_k^* \geq 0$, which implies

$$A_k = -A_k^* \qquad \forall k \in \underline{K} \tag{5}$$

since $K_k = jA_k$. We now choose a sufficiently small $\varepsilon > 0$. As $M(s)$ is an NI system,

$$
\begin{aligned}
& j[M(j\omega) - M(j\omega)^*] \geq 0 \ \ \forall \omega \in \{\omega : 0 < |\omega - \omega_k| < \varepsilon,\ \forall k \in \underline{K}\} \\
\Leftrightarrow\ & j\left[\left(M_k(j\omega) + \frac{A_k}{j(\omega - \omega_k)}\right) - \left(M_k(j\omega) + \frac{A_k}{j(\omega - \omega_k)}\right)^*\right] \geq 0 \\
& \forall \omega \in \{\omega : 0 < |\omega - \omega_k| < \varepsilon,\ \forall k \in \underline{K}\} \\
\Leftrightarrow\ & j[M_k(j\omega) - M_k(j\omega)^*] + \frac{A_k + A_k^*}{(\omega - \omega_k)} \geq 0 \\
& \forall \omega \in \{\omega : 0 < |\omega - \omega_k| < \varepsilon,\ \forall k \in \underline{K}\} \\
\Rightarrow\ & j[M_k(j\omega_k) - M_k(j\omega_k)^*] \geq 0 \quad \forall k \in \underline{K}
\end{aligned} \tag{6}
$$

since $A_k + A_k^* = 0 \quad \forall k \in \underline{K}$ from (5). Now assumption (3) implies $\det[N(j\omega_k)] \neq 0 \quad \forall k \in \underline{K}$ via Lemma 3 and ensures

$$(jN(j\omega_k))^{-1} + (jN(j\omega_k))^{-*} > 0 \qquad \forall k \in \underline{K}. \tag{7}$$

Inequalities (6) and (7) together yield

$$
\begin{aligned}
& [(jN(j\omega_k))^{-1} + (jM_k(j\omega_k))] \\
& \quad + [(jN(j\omega_k))^{-1} + (jM_k(j\omega_k))]^* > 0 \qquad \forall k \in \underline{K}.
\end{aligned} \tag{8}
$$

Therefore, $\det[(jN(j\omega_k))^{-1} + (jM_k(j\omega_k))] \neq 0\ \forall k \in \underline{K}$. But, we have to show that $[I - M(s)N(s)]$ has no transmission zero at $s = j\omega_k$ for any $k \in \underline{K}$, which is equivalent to $[jN(s)]^{-1} + [jM(s)]$ having no transmission zero at $s = j\omega_k$ for any $k \in \underline{K}$. We show this via contradiction. Suppose $s = j\omega_k$ is a transmission zero of $[jN(s)]^{-1} + [jM(s)]$ for some $k \in \underline{K}$. Then, there exists a non-zero vector $y_k \in \mathbb{C}^m$ such that $[(jN(s))^{-1} + (jM(s))]y_k = 0$ at $s = j\omega_k$. Expanding the above equation, we have

$$[jN(s)]^{-1}y_k + [jM_k(s)]y_k = \frac{-jA_k\, y_k}{s - j\omega_k} \tag{9}$$

when $s = j\omega$ with $0 < |\omega - \omega_k| < \varepsilon$. Now, pre- and post-multiplying the left-hand side of (8) by y_k^* and y_k and then substituting (9) into the expression when $s \to j\omega_k$, we obtain

$$
\begin{aligned}
& y_k^*\Big[(jN(j\omega_k))^{-1} + (jM_k(j\omega_k))\Big] \\
& \quad + \Big[(jN(j\omega_k))^{-1} + (jM_k(j\omega_k))\Big]^* y_k \\
& = \lim_{\omega \to \omega_k} y_k^*\left[\frac{-jA_k}{j\omega - j\omega_k} + \frac{jA_k^*}{-j\omega + j\omega_k}\right]y_k \\
& = \lim_{\omega \to \omega_k} -y_k^*\left[\frac{(A_k + A_k^*)}{\omega - \omega_k}\right]y_k = 0,
\end{aligned}
$$

since $A_k + A_k^* = 0\ \forall k \in \underline{K}$ via (5). Therefore, it is clear that (9) violates (8), which implies that there does not exist any such $y_k \in \mathbb{C}^m$ such that $[(jN(s))^{-1} + (jM(s))]y_k = 0$ at $s = j\omega_k$ for any $k \in \underline{K}$. Therefore, $[I - M(s)N(s)]$ does not have any transmission zero at $s = j\omega_k$ for any $k \in \underline{K}$.

Similarly, it can be established that $[I - M(s)N(s)]$ does not have any transmission zero at $s = j\omega_l$ for any $l \in \underline{L}$ exploiting supposition (4) of this lemma.

Part 2 ($\omega \in (0,\infty)\backslash(\Omega_m \cup \Omega_n)$): For convenience, we define the following sets of frequencies:

$$
\begin{aligned}
\mathbb{Z}_m &= \{\omega \in (0,\infty)\backslash(\Omega_m \cup \Omega_n) : \det\big[M(j\omega) - M(j\omega)^*\big] \neq 0\}; \\
\mathbb{Z}_n &= \{\omega \in (0,\infty)\backslash(\Omega_m \cup \Omega_n) : \det\big[N(j\omega) - N(j\omega)^*\big] \neq 0\}; \\
\mathbb{Z}_{mn} &= \{\omega \in (0,\infty)\backslash(\Omega_m \cup \Omega_n) : \det\big[M(j\omega) - M(j\omega)^*\big] \neq 0 \\
&\qquad \text{and } \det\big[N(j\omega) - N(j\omega)^*\big] \neq 0\},
\end{aligned}
$$

$$\mathbb{Z}_{\overline{mn}} = \{\omega \in (0,\infty)\backslash(\Omega_m \cup \Omega_n) : \det\big[M(j\omega) - M(j\omega)^*\big] = 0 \text{ and } \det\big[N(j\omega) - N(j\omega)^*\big] = 0\}.$$

It is evident that $\omega \in (0,\infty)\backslash(\Omega_m \cup \Omega_n) = \mathbb{Z}_m \cup \mathbb{Z}_n \cup \mathbb{Z}_{mn} \cup \mathbb{Z}_{\overline{mn}}$. The assumption that there does not exist any $\omega \in (0,\infty)\backslash(\Omega_m \cup \Omega_n)$ such that $\det[M(j\omega) - M(j\omega)^*] = 0$ and $\det[N(j\omega) - N(j\omega)^*] = 0$ signifies that $\mathbb{Z}_{\overline{mn}} = \emptyset$. Since $\det[M(j\omega) - M(j\omega)^*] \neq 0\ \forall\omega \in \mathbb{Z}_m$, it implies $j[M(j\omega) - M(j\omega)^*] > 0\ \forall\omega \in \mathbb{Z}_m$ (due to $M(s)$ being NI) which, in turn, implies $\det[M(j\omega)] \neq 0\ \forall\omega \in \mathbb{Z}_m$ via Lemma 3. Therefore, for all $\omega \in \mathbb{Z}_m$, we have

$$\det[(jM(j\omega))]\det[(jM(j\omega))^{-1} + jN(j\omega)] \neq 0 \tag{10}$$

which implies

$$\det[I - M(j\omega)N(j\omega)] \neq 0 \qquad \forall\omega \in \mathbb{Z}_m. \tag{11}$$

Similarly, it can be shown that for all $\omega \in \mathbb{Z}_n$, $j[N(j\omega) - N(j\omega)^*] > 0$ and hence, $\det[N(j\omega)] \neq 0$ via Lemma 3. Therefore, for all $\omega \in \mathbb{Z}_n$, we have

$$\det[(jN(j\omega))]\det[(jN(j\omega))^{-1} + jM(j\omega)] \neq 0 \tag{12}$$

which implies

$$\det[I - M(j\omega)N(j\omega)] \neq 0 \qquad \forall\omega \in \mathbb{Z}_n. \tag{13}$$

Since $\mathbb{Z}_{mn} = \mathbb{Z}_m \cap \mathbb{Z}_n$, it follows that

$$\det[I - M(j\omega)N(j\omega)] \neq 0 \qquad \forall\omega \in \mathbb{Z}_{mn}. \tag{14}$$

Combining the results (11), (13) and (14), it can be concluded that $\det[I - M(j\omega)N(j\omega)] \neq 0\ \forall\omega \in (0,\infty)\backslash(\Omega_m \cup \Omega_n)$ which is equivalent to $[I - M(j\omega)N(j\omega)]$ having no transmission zeros on the $j\omega$-axis for any $\omega \in (0,\infty)\backslash(\Omega_m \cup \Omega_n)$.

Hence, Part 1 and Part 2 jointly prove that $[I - M(s)N(s)]$ does not have any transmission zero at $s = j\omega$ for any $\omega \in (0,\infty)$ for Case I.

Case II: Suppose $M(s)$ is a marginally stable NI system where $[M(s) - M^{\sim}(s)]$ does not have full normal rank, and $N(s)$ is a stable NI system. Therefore, $\det[M(j\omega) - M(j\omega)^*] \equiv 0$ $\forall\omega \in (0,\infty)\backslash\Omega_m$. This particular case includes the Lossless NI systems (as in [25]). One can assure that Part 1 of Case I (i.e., (8) holds) still remains applicable in this situation since $N(s)$ is stable. However, Part 2 of Case I is only partially fulfilled in this case, since $\mathbb{Z}_{\overline{mn}} \neq \emptyset$. The condition (12) does not hold for the finite set of frequencies $\omega_0 \in \{\omega \in (0,\infty)\backslash(\Omega_m \cup \Omega_n) : \det[N(j\omega) - N(j\omega)^*] = 0\}$. For such frequencies, exploiting the last assumption of Lemma 4, together with (12) for the rest of the frequencies when $\det[N(j\omega) - N(j\omega)^*] \neq 0$, we can directly arrive at the condition $\det[I - M(j\omega)N(j\omega)] \neq 0\ \forall\omega \in (0,\infty)\backslash(\Omega_m \cup \Omega_n)$. This, hence, proves that $[I - M(s)N(s)]$ does not have any transmission zero at $s = j\omega$ for any $\omega \in (0,\infty)$. The same arguments apply for swapped conditions on $M(s)$ and $N(s)$.

Case III: Now, suppose $M(s)$ is a marginally stable NI system with $[M(s) - M^{\sim}(s)]$ having full normal rank, and $N(s)$ is a stable NI system. It can be readily established that $[I - M(s)N(s)]$ does not have any transmission zero on the $j\omega$-axis for any $\omega \in (0,\infty)$ via an argument similar to Part 2 of Case I upon using the assumption that $\nexists\,\omega \in (0,\infty)\backslash(\Omega_m \cup \Omega_n)$ such that $\det[M(j\omega) - M(j\omega)^*] = 0$ and $\det[N(j\omega) - N(j\omega)^*] = 0$.

Case IV: Suppose that $M(s)$ and $N(s)$ are both stable NI systems. It can be readily established that $\det[I - M(j\omega)N(j\omega)] \neq 0\ \forall\omega \in (0,\infty)$, since $M(s) \in \mathcal{RH}_\infty$, $N(s) \in \mathcal{RH}_\infty$, and via supposition $\nexists\,\omega \in (0,\infty)\backslash(\Omega_m \cup \Omega_n)$ such that $\det[M(j\omega) - M(j\omega)^*] = 0$ and $\det[N(j\omega) - N(j\omega)^*] = 0$.

Case I, Case II, Case III and Case IV complement each other to prove the lemma by addressing all possible combinations of the NI systems $M(s)$ and $N(s)$. ■

Remark 1: Case III of Lemma 4, when specialized to a NI–SNI interconnection, captures the result of [26, Lemma 6], and also corrects the proof of the latter to reflect the fact that the NI system (say, $M(s)$) allows multiple non-repeated poles on the $j\omega$-axis for $\omega \in (0,\infty)$. In [26, Lemma 6], $M(s)$ is decomposed as $M_1(s) + \frac{-jA}{s-j\omega_0}$ with $A = A^* \geq 0$ and the proof extensively relies on the assumption that A does always have a kernel. But, A may not have a kernel, e.g., $M(s) = \frac{1}{s^2+4} = M_1(s) + \frac{-j\frac{1}{4}}{s-j2}$ gives $A = \frac{1}{4} > 0$. In such cases, proof of [26, Lemma 6] does not hold. On the contrary, in the present approach, the proof does not depend on the existence of the kernel of A.

Also note that when $M(s)$ is a stable NI system and $N(s)$ is an SNI system, Lemma 4 of this letter specializes to the [1, Lemma 4], which can be readily comprehended from Case III of the proof of Lemma 4.

With the help of Lemma 4, the following theorem proves the internal stability condition for a positive feedback interconnection of two NI systems with possible poles on the $j\omega$-axis excluding the origin (i.e., both systems can be *non-strict* NI).

Theorem 2: Let $M(s)$ and $N(s)$ be two NI systems with possible poles on the $j\omega$-axis excluding the origin. Let $M(s)$ and $N(s)$ satisfy the suppositions of Lemma 4. Then, the positive feedback interconnection of $M(s)$ and $N(s)$ (shown in Fig. 1) is internally stable if and only if

$$\det[I - M(\infty)N(\infty)] \neq 0, \tag{15a}$$

$$\lambda_{\max}\Big[[I - M(\infty)N(\infty)]^{-1}(M(\infty)N(0) - I)\Big] < 0, \tag{15b}$$

$$\lambda_{\max}\Big[[I - N(0)M(\infty)]^{-1}(N(0)M(0) - I)\Big] < 0. \tag{15c}$$

Proof: Let $\left[\begin{array}{c|c} A_1 & B_1 \\ \hline C_1 & D_1 \end{array}\right]$ and $\left[\begin{array}{c|c} A_2 & B_2 \\ \hline C_2 & D_2 \end{array}\right]$ be the minimal state-space realizations of $M(s)$ and $N(s)$ respectively with $\det[A_1] \neq 0$, $\det[A_2] \neq 0$, $D_1 = D_1^\top$ and $D_2 = D_2^\top$. Applying the NI lemma (Lemma 1), there exist real matrices $Y_1 = Y_1^\top > 0$ and $Y_2 = Y_2^\top > 0$ such that $M(s)$ and $N(s)$ satisfy

$$A_1Y_1 + Y_1A_1^\top \leq 0 \quad \text{and} \quad B_1 + A_1Y_1C_1^\top = 0;$$
$$A_2Y_2 + Y_2A_2^\top \leq 0 \quad \text{and} \quad B_2 + A_2Y_2C_2^\top = 0.$$

This theorem will be proved in four parts – Cases I, II, III, and IV (following similar grouping as in Lemma 4).

Case I: Suppose $M(s)$ and $N(s)$ have, respectively, $K \in \mathbb{N}$ and $L \in \mathbb{N}$ non-repeated poles on $j\omega$-axis for $\omega \in (0,\infty)$. Define the sets $\underline{K} = \{1, 2, \ldots, K\}$ and $\underline{L} = \{1, 2, \ldots, L\}$. Since both $M(s)$ and $N(s)$ do not have any poles in the open right-half plane or at the origin and, via supposition, $M(s)N(s)$ does not have any pole-zero cancellation on the $j\omega$-axis for any $\omega \in (0,\infty)$, it can be easily seen that $M(s)N(s)$ does not cause any pole-zero cancellation in the entire closed right-half plane. Note that $[M(s) - M^{\sim}(s)]$ and $[N(s) - N^{\sim}(s)]$ possess full normal rank, as explained in Lemma 4.

To proceed with the proof, we define the following matrices $U = I - D_1D_2$, $V = I - D_2D_1$ and $\Phi = \begin{bmatrix} A_1Y_1 & 0 \\ 0 & A_2Y_2 \end{bmatrix}$. Both U^{-1} and V^{-1} exist as the positive feedback interconnection is considered to be well-posed. Φ is non-singular since all of its constituent matrices are non-singular. Now, the positive feedback interconnection of $M(s)$ and $N(s)$ in Fig. 1 is internally stable if and only if $\det[I - M(\infty)N(\infty)] \neq 0$ and $[I - M(s)N(s)]^{-1} = \left[\begin{array}{c|c} A_{cl} & B_{cl} \\ \hline C_{cl} & D_{cl} \end{array}\right] \in \mathscr{RH}_{\infty}^{m\times m}$ where $A_{cl} = \begin{bmatrix} A_1 + B_1D_2U^{-1}C_1 & B_1V^{-1}C_2 \\ B_2U^{-1}C_1 & A_2 + B_2U^{-1}D_1C_2 \end{bmatrix}$, $B_{cl} = \begin{bmatrix} B_1D_2U^{-1} \\ B_2U^{-1} \end{bmatrix}$, $C_{cl} = \begin{bmatrix} U^{-1}C_1 & U^{-1}D_1C_2 \end{bmatrix}$ and $D_{cl} = U^{-1}$. This equivalence holds due to [28, Th. 5.7] under the supposition that $M(s)N(s)$ does not have any pole-zero cancellation in the entire closed right-half plane. We then utilize the relationship $A_{cl} = \Phi\mathscr{T}$, where the symmetric matrix

$$\mathscr{T} = \begin{bmatrix} Y_1^{-1} - C_1^{\top}D_2U^{-1}C_1 & -C_1^{\top}V^{-1}C_2 \\ -C_2^{\top}U^{-1}C_1 & Y_2^{-1} - C_2^{\top}U^{-1}D_1C_2 \end{bmatrix}$$

plays the role of a Lyapunov candidate matrix in order to prove the Hurwitzness of A_{cl}. Now, proceeding along the lines of proof of [26, Th. 9], we obtain that A_{cl} is Hurwitz if and only if $\mathscr{T} > 0$, under the assumption that $\det[I - M(\infty)N(\infty)] \neq 0$. It can further be derived via a sequence of matrix manipulations that $\mathscr{T} > 0$ if and only if the set of conditions (15) holds. This, in turn, guarantees the internal stability of the positive feedback interconnection of $M(s)$ and $N(s)$. Lemma 4 has been exploited[2] in this proof to deal with the fact that now $M(s)$ and $N(s)$ both allow non-repeated poles on the $j\omega$-axis for $\omega \in (0, \infty)$, contrary to all the earlier internal stability results available in the NI literature [3], [9], [25], [26]. This completes the proof for Case I of Theorem 2.

Case II: Suppose $M(s)$ is a marginally stable NI system for which $[M(s) - M^{\sim}(s)]$ does not have full normal rank, and $N(s)$ is a stable NI system. Since $N(s)$ does not have any pole in the closed right-half plane and $M(s)N(s)$ does not give rise to any pole-zero cancellation on the $j\omega$-axis $\forall\omega \in \Omega_m$ due to satisfying the condition (3), it confirms $M(s)N(s)$ does not have any pole-zero cancellation in the entire closed right-half plane. Now, following Case I of the current proof and utilizing Case II of Lemma 4, it can be readily established that the positive feedback interconnection of $M(s)$ and $N(s)$ is internally stable if and only if (15) are satisfied.

Case III: Now, suppose $M(s)$ is a marginally stable NI system with $[M(s) - M^{\sim}(s)]$ having full normal rank, and $N(s)$ is a stable NI system. We can readily confirm via Case II that $M(s)N(s)$ does not have any pole-zero cancellation in the entire closed right-half plane. The internal stability of $M(s)$ and $N(s)$ follows along the lines of Case I and Case II taking the help from Case III of Lemma 4.

Case IV: Suppose finally that both $M(s)$ and $N(s)$ are stable NI systems. Hence, the requirement that $M(s)N(s)$ should not have any pole-zero cancellation in the entire closed right-half plane, is trivially satisfied. Then, following similar arguments as used in Case III above and in Case IV of Lemma 4, the positive feedback interconnection of $M(s)$ and $N(s)$ can be guaranteed to be internally stable.

Thus, Case I, Case II, Case III and Case IV together complete the proof. ■

Remark 2: Case II of Theorem 2 offers a valuable result that a positive feedback interconnection of a Lossless NI system $M(s)$ and a *non-strict*, stable NI system $N(s)$ can be internally stable. This finding contrasts with [25, Th. 2] which says that (using NI theory) a Lossless NI system can be stabilized only by an SNI system in a positive feedback loop.

Remark 3: Case III reveals that a positive feedback interconnection of a *marginally stable* NI system without poles at the origin and a *non-strict*, stable NI system can still be internally stable, satisfying the same set of conditions (15). This is a significant finding against the existing notion that to ensure internal stability of a positive feedback interconnection of two NI systems, at least one of the systems must be SNI [26, Th. 9]. Furthermore, when $N(s)$ is restricted to be SNI, Case III captures the internal stability theorem of a NI–SNI interconnection derived in [3, Th. 1], [26, Th. 9].

Remark 4: Case IV emphasizes another significant result that a positive feedback interconnection of two *non-strict*, stable NI systems can be internally stable without imposing the constraint that one of the systems must be SNI, unlike [1, Th. 5]. Moreover, when $N(s)$ is assumed to be SNI and the assumptions $M(\infty)N(\infty) = 0$ and $N(\infty) \geq 0$ are back in place, Case IV directly captures the internal stability theorem of a stable NI–SNI interconnection introduced in [1].

IV. Numerical Examples

This section presents two numerical examples – Example 1 signifies that a *non-strict*, stable NI system can *robustly* stabilize an uncertain, Lossless NI system in a positive feedback loop; while Example 2 illustrates the robust stability of a positive feedback interconnection of two MIMO *marginally stable* NI systems. These examples will highlight the usefulness and the potential significance of the proposed result over the existing NI–SNI internal stability results [1], [3], [25], [26].

Example 1: Consider a positive feedback loop containing a Lossless NI system $M(s) = \dfrac{2k(-s^2 - 10)}{2s^2 + 20 + k}$, where $k > 0$ is an uncertain parameter, and a *non-strict* stable NI system $N(s) = \dfrac{2s^2 + s + 1}{(s+1)(2s+1)(s^2 + 2s + 5)}$. $M(s)$ has a purely imaginary pole-pair at $s = \pm j\sqrt{\frac{20+k}{2}}$ and zeros at $s = \pm j\sqrt{10}$; while $\det[N(j\omega_0) - N(j\omega_0)^*] = 0$ at $\omega_0 = 1$ rad/s. With careful observation, we confirm that (i) $M(s)N(s)$ does not have any pole-zero cancellation in the entire closed right-half plane $\forall k > 0$, (ii) $j[N(j\omega) - N(j\omega)^*] > 0$ at $\omega = \sqrt{\frac{20+k}{2}}$ rad/s $\forall k > 0$, and (iii) $\det[I - M(j\omega_0)N(j\omega_0)] = 1 + \frac{1.8k}{18+k} \neq 0$ for any $k > 0$. We observe that Case II of Theorem 2 applies to the present case as $M(s)$ is a Lossless NI system and $N(s)$ is a non-strict, stable NI system. Since $M(\infty)N(\infty) = 0$ and $N(\infty) = 0$, the set of conditions (15) boils down to the more elegant and popular *DC loop gain* condition: $N(0)M(0) < 1$, which is satisfied $\forall k > 0$ since $N(0)M(0) = -\frac{4k}{20+k} < 1$ $\forall k > 0$. This, hence,

[2]Note that [26, Lemma 6] is not applicable in the proof of Theorem 2 because none of the systems in the interconnection is SNI.

guarantees that the positive feedback interconnection of $M(s)$ and $N(s)$ is *robustly* $(\forall k > 0)$ internally stable.

Example 2: Consider a positive feedback interconnection of two MIMO (2×2) *marginally stable* NI transfer function matrices $M(s) = \begin{bmatrix} \frac{s+a}{s^2+10s+25} & \frac{-0.5}{s+5} \\ \frac{-0.4s-0.5}{s^2+6s+5} & \frac{0.7s^3+4s^2+14.8s+18}{s^4+6s^3+9s^2+24s+20} \end{bmatrix}$ and

$$N(s) = \begin{bmatrix} \frac{0.5s^2+5s+29.5}{s^3+5s^2+9s+45} & \frac{-0.5}{s+5} \\ \frac{-0.4s-0.5}{s^2+6s+5} & \frac{2s^2+bs+17}{s^3+12s^2+41s+30} \end{bmatrix},$$

where a and b are two uncertain parameters varying in the intervals $[10, 30]$ and $[5, 20]$, respectively. Both the systems continue to satisfy the NI property even with the uncertain parameters. Each system has purely imaginary poles: $M(s)$ at $s = \pm j2$ and $N(s)$ at $s = \pm j3$. Hence, although both being NI, none of them satisfy SNI property. However, one can readily test that $M(s)$ and $N(s)$ satisfy all the suppositions of Theorem 2 for the prescribed range of values of the uncertain parameters a and b. Finally, upon inspection of the DC loop gain matrix $N(0)M(0)$, it can be easily verified that the condition $\lambda_{\max}[N(0)M(0)] = \lambda_{\max}\begin{pmatrix} \frac{59a}{2250} + \frac{1}{100} & -\frac{7}{45} \\ -\frac{a}{250} - \frac{17}{300} & \frac{13}{25} \end{pmatrix} < 1$ always gets satisfied for any value of a in the interval $[10, 30]$. This guarantees that the positive feedback interconnection of $M(s)$ and $N(s)$ is *robustly* internally stable via Theorem 2.

V. Conclusion

This brief derives an internal stability result for positive feedback interconnections of NI systems, allowing the cases when both the systems may have poles on the imaginary-axis (excluding the origin). The proposed theorem, therefore, generalizes the existing internal stability results for an NI interconnection since it does not ask for any of the systems in the interconnection to satisfy *strict* NI (i.e., SNI) property, which has been a must-to-hold condition in the NI literature. The proposed stability theorem also captures the available internal stability results of a NI–SNI interconnection without poles at the origin as special cases. This generalization would widen the applicability of NI stability theory and may simplify the NI synthesis methodologies.

Acknowledgment

The authors would like to convey sincere thanks to Prof Alexander Lanzon, Dept. of Electrical and Electronic Engg., The University of Manchester, U.K., and Dr Sourav Patra, Dept. of Electrical Engg., Indian Institute of Technology Kharagpur, India, for their kind support and encouragement.

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